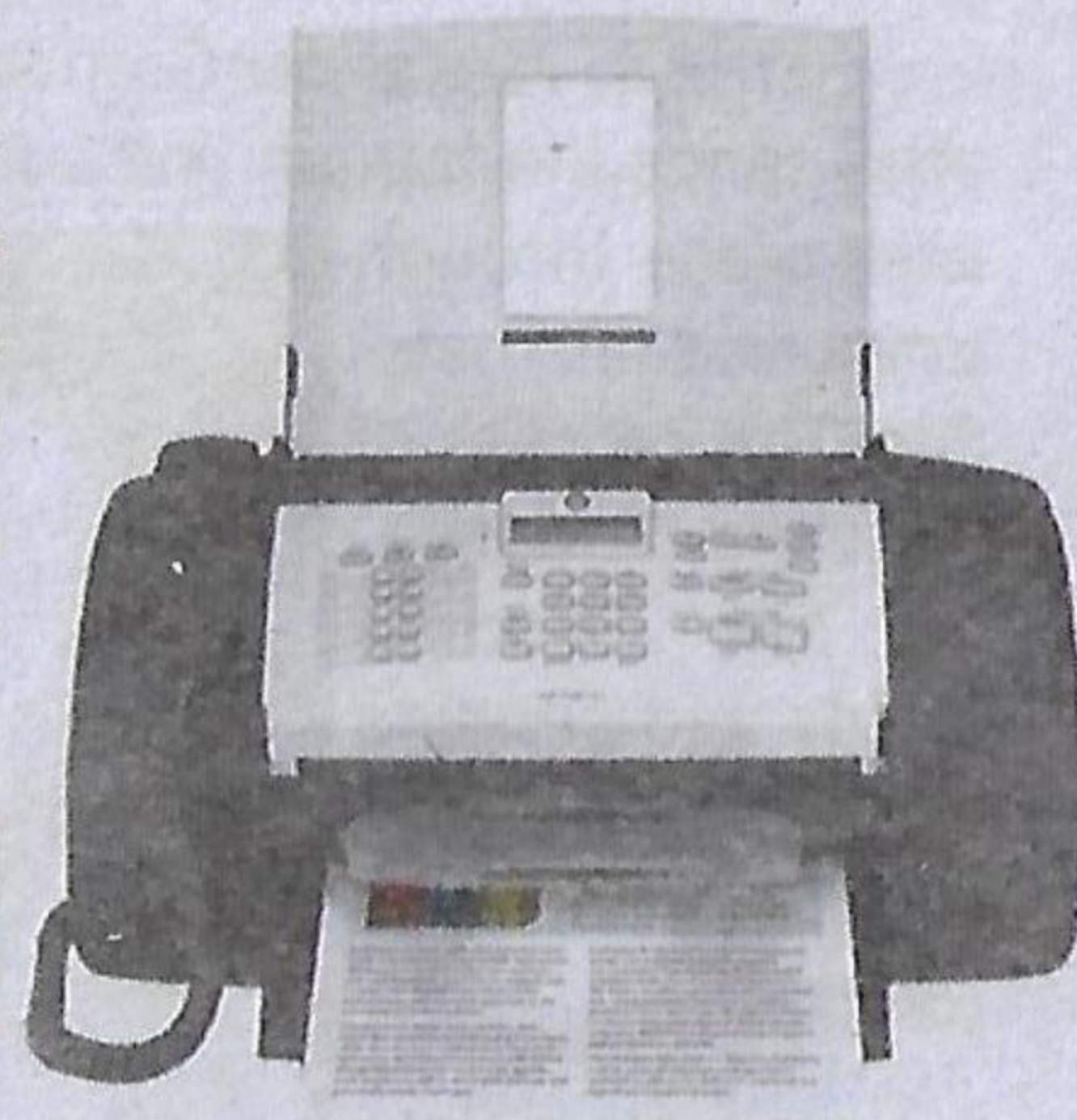


from the telephone company. Telephone networks modulate voice communications over these wires. A complex system of network switches maintains the telephone links between callers. Telephone networks also use microwave relay stations to send calls from place to place on the ground. Satellites are used by telephone networks to transmit telephone calls across continents and oceans.

Teletype, Telex, and Facsimile Transmission (Fax)

Teletype, Telex, and Facsimile (Fax) transmission are all methods for transmitting text rather than sounds. These text delivery systems were evolved from the telegraph. Teletype and telex systems still exist, but they have been largely replaced by Facsimile (Fax) machines, which are inexpensive and can better be operated over the existing telephone network. The Internet provides an even more inexpensive and convenient option. The Teletype, essentially a printing telegraph, is primarily a point-to-multipoint system for sending text. The Teletype converts the same pulses used by telegraphs into letters and numbers and then prints out readable text. It was often used by news media organizations to provide newspaper stories and stock market data to subscribers. Telex is primarily a point-to-point system that uses a keyboard to transmit typed text over telephone lines to similar terminals situated at individual company locations.

Facsimile (Fax) transmission now provides a cheaper and easier way to transmit text and graphics over long distances. Fax machines contain an optical scanner that converts text and graphics into digital, or machine-readable codes. This coded information is sent over ordinary analog telephone lines through the use of a modem included in the fax machine. The receiving fax machine's modem demodulates the signal and sends it to a printer also contained in the fax machine.



Facsimile (Fax) Machine

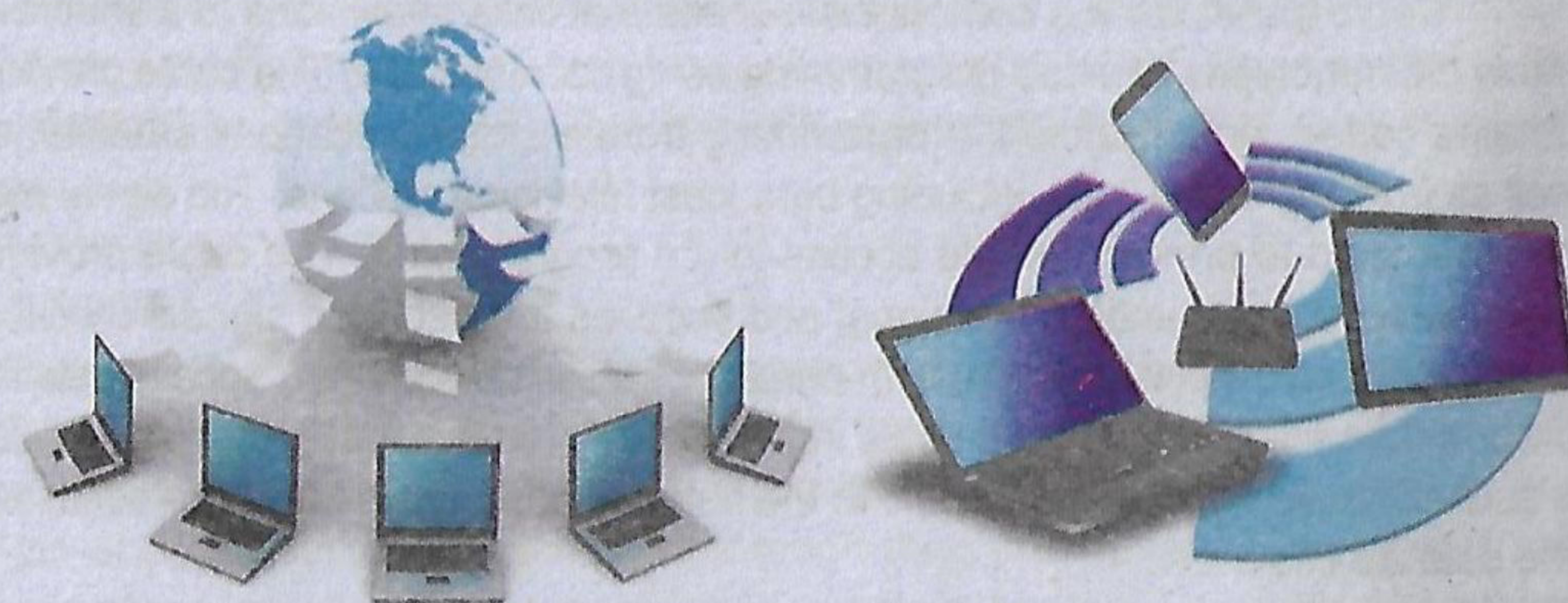
Computer

In today's developed world computer is an important means of communication. This is used for different purposes on land and space. For reserving seats the air companies make use of the computers. Through computers the flight schedule can be known. This also helps in air traffic control. Through computers the pilots are

guided. These are also used on a large scale in space. For sending satellites in space and for achieving the various objectives the entire system needs to be taken care of. Many decisions have to be taken immediately and on time, which are only possible through computers.

Personal computers use telecommunications to provide a transmission link for the delivery of audio, video, text, software, and multimedia services. Currently, the delivery of most of these audio, video, and text services occurs over existing telephone connections using the Internet.

Electronic mail, or e-mail, is a key attraction of the Internet and a common form of computer telecommunications. E-mail is a text-based message delivery system that allows information such as typed messages and multimedia to be sent to individual computer users. Local e-mail messages (within a building or a company) typically reach addressees by travelling through wire-based internal networks. E-mail that must travel across town or across a country to reach the final destination usually travels through the telephone network. Other computer telecommunication technologies that businessmen frequently use, include automated banking terminals and devices for credit card transactions that bill charges directly to a customer's bank account.



Computer Networking

Radio and T.V

In 1895 Marconi invented the radio. He invented a system with the help of which electrical waves without wires could be sent in the air. This led to the invention of radio. Radio is used mostly as a public medium, sending commercial broadcasts from a transmitter to anyone with a radio receiver within its range, so it is known as a point-to-multipoint medium. However, radio can also be used for private point-to-point transmissions. Two-way radios, cordless telephones, and cellular radio telephones are common examples of transceivers, which are devices that can both transmit and receive point-to-point messages.

Television is primarily a public broadcasting medium, using point-to-multipoint technology that is broadcast to any user within range of the transmitter. Televisions transmit news and information, as well as entertainment. Commercial television is broadcast over very high frequency (VHF) and ultrahigh frequency (UHF) radio waves and can be received by any television set within the range of the transmitter. Televisions have also been used for point-to-point, two-way telecommunications. Teleconferencing, in which a television picture links two physically separated parties, is a convenient way for businesspeople to meet and communicate without the expense or inconvenience of travel. Video cameras on computers now allow personal computer users to teleconference over the Internet. Videophones, which use tiny video cameras and rely on satellite technology, can also send private or public television images and have been used in news reporting in remote locations.

Cable television is a commercial service that links televisions to a source of many different types of video programming using coaxial cable. The cable provider obtains coded, or scrambled, programming from a communications satellite, as well as from terrestrial links, including broadcast television stations. The signal may be scrambled to prevent unpaid access to the programming. The cable provider electronically unscrambles the signal and supplies the decoded signals by cable to subscribers. Television users with personal satellite dishes can access satellite programming directly without a cable installation. Personal satellite dishes are also a subscriber service. Fees are paid to the network operator in return for access to the satellite channels.

Means of Transportation

Means of transportation serve as the basis of communication between the buyers and the sellers. Means of communication reduce the distance between markets and place of production. For the exchange of trade goods, appropriate means of

transportation are essential. Until and unless the means of transportation of a country are not fully developed the agriculture and industrial produce of any country gets wasted. Without it trade and economic development is not possible. If these goods reach the consumers easily then their demand and value increases. Production cannot reach consumers without proper means of transportation. Therefore, the success of trade is dependent on the quality of the means of transportation.

There are three kinds of the means of transportation i.e., land, air, and sea.

Means of Transportation Via Land

This includes the system of rail and road. In most of the countries of the world the means of transportation are poor and inadequate whereas in some of the developed parts these means are most efficient and modern. The developed nations of the world have a well-connected system of rail and road. In the cities and the villages there is a good network of roads. There are wide roads in Germany, Britain, America and Australia, which are called highways or motorways. These are made especially for fast and heavy traffic. In the developing nations there are a few roads of this kind, which connect important cities. In the developed nations train is the most important and cheap means of transportation. The best system of railway is present in Japan, America, Canada, Russia and Australia. In these countries railway carriages are mostly used for transporting heavy goods like raw material, coal, industrial goods, construction wood to far flung areas. Almost all the European countries are linked to each other through railway lines. Japan has an efficient and modern system of railways, which is very famous in the world. Asian countries like Pakistan, India, Indonesia, Thailand and Singapore also have a network of railway lines.

Means of Transportation Via Sea

Two third area of the Earth is covered with water. This is a great blessing of Allah Almighty, which has many benefits for mankind. The greatest benefit is that it is used for various means of transportation as the majority of the countries of the world are linked to each other through seas and oceans. The biggest sea routes of the world include oceans seas, rivers and lakes. The route of northern Pacific Ocean is the most important trade route of this ocean. Majority of the sea ships carrying goods and passengers pass through it. The other important trade route is through the Pacific Ocean and Mediterranean sea. Through this route minerals and agricultural produce and construction wood is taken to Europe and

industrial goods come from Europe to countries located in the east. The third largest sea route includes the trade route between North Atlantic Ocean and Far Eastern countries and other Asian countries, which trade with the US. In the ancient times trade activities were carried out through the rivers. In most of the countries of the world shipping is done through rivers and the trade within countries is also carried out through rivers. The important rivers of the world on which trade is carried out include River Thames, and the estuary of Humber in Northeast England, formed by the Ouse and Trent rivers in Britain, River Danube of Germany, River Volga (Russian name) or Volga (English name) of Russia. A Canadian River is joined to big lakes through smaller lakes and through these lakes ships can have direct access to the Pacific Ocean. In the United States of America, River Mississippi is also fit for shipping. Shipping is also possible in the Chang Jiang River of China.

Suez Canal is the most important water highway of the world, which joins two important seas of the world Mediterranean Sea and Red Sea. The Suez Canal is situated in Egypt. This has been made by cutting the narrow land route, which joins Asia to Africa. Before the construction of Suez Canal, coming from Europe to the Asian countries was difficult and an entire round of Africa had to be taken for doing so. Now this canal has reduced the distance considerably.

Panama Canal has been constructed in Panama in Central America. This joins two oceans Pacific Ocean and Atlantic Ocean through Caribbean Sea. USA, Canada and others countries are benefited by this canal.

Means of Transportation Via Air

The air routes have transformed the world into a global village. People living in far-flung areas can cover great distances in a short time. Even though air transportation is expensive but businessmen and people travelling to far flung areas have felt its importance and necessity.

Among the important world air routes, the following are famous:

- + In America there are many national and international air routes. On the eastern coast New York is the biggest airport.
- + The capitals of almost all the European countries are important air centres. Many air routes are linked to western Africa through Europe. Many air routes link the North African countries.



Quaid-i-Azam International Airport, Karachi.

- + In Middle East the important air routes of Dubai, Kuwait and Bahrain connect America directly to the far eastern countries.
- + Australia is also significant from the point of view of air transportation. Domestic and international flights go from here. Sydney is also an important air centre.
- + Similarly the air routes of Pakistan and India in South Asia are also very important.

Trade Between Pakistan and the World

Import and Export:

Any country of the world inspite of being highly developed cannot be self-sufficient in meeting all its needs and requirements. Every country has to buy different commodities from other countries and has to sell its surplus things to those countries. This process takes place within the country as well as outside the country. The buying and selling and exchange of goods is called TRADE. The trade, which takes place within the country (between the different parts of the country), is called INTERNAL or DOMESTIC trade and the trade, which takes place between two countries, is called EXTERNAL or INTERNATIONAL trade. Trade has two dimensions one is buying and the other is selling. Those goods, which are sold to other countries, are called EXPORTED goods and this trade is called export. Contrary to this those goods, which are bought from other countries into the country, are called IMPORTED items and forms its import. All countries have two types of trade one is import and

the other is export. Usually, trade goods are purchased on cash. But sometimes goods are also exchanged for goods. This kind of trade is called "BARTER TRADE".

Foreign Exchange

The exchange items are generally bought on cash. As the currency of every country is different therefore exchange takes place on the basis of some mutually agreed currency. In Pakistan trade mostly takes place in American dollars or British pounds. The money, which is obtained from external trade, is called FOREIGN EXCHANGE. If any country produces any commodity exactly according to its needs that country is called SELF-SUFFICIENT in that particular commodity. For example, if Pakistan produces sufficient wheat for its requirement then Pakistan will be called self-sufficient in wheat. In any country if goods are produced less than the requirements of the country then that country is called low producer of that commodity and if that country produces a commodity in abundance then it is called a high producer of that commodity.

Means of Transportation of Goods and Trade Routes

For the exchange of trade goods, means of transportation are very essential. Until and unless these means of transportation are not modern and effective no country's agriculture or industrial development has any importance from the point of view of trade and economics. If these goods reach the consumers easily then their utilization and importance increases. Transportation of goods from the place of production to the place of exchange is not possible without proper system of transportation. Therefore, success of trade depends on the quality of transportation of goods. There are three means by which goods can be transported viz a viz land, sea and air.

Land Routes

As compared to the ancient times, today's means of transportation are better and efficient. There are many good roads on which different vehicles ply easily. There are railway tracks also and goods are sent to suitable places for internal and external trade. For internal trade excluding the rugged areas of Balochistan and N.W.F.P in the entire country there is a network of railway tracks and roads. Trade also takes place via road with the neighboring countries of India, Afghanistan and Iran. According to this agreement made by the Economic Co-operation Organization the planning for the construction of a highway from Karachi to Iran and Turkey has been completed. The other important trade land route of Pakistan is the Karakoram Highway, which connects Pakistan with the Peoples Republic of China.

Sea Routes

The external trade of Pakistan mostly takes place through sea routes for which the seaport of Karachi is used. From Pakistan, trade goods are sent to the eastern countries of India, Sri Lanka, Bangladesh, Malaysia, Indonesia, China and Japan. The other route from Persian Gulf and Suez Canal goes to Middle East, North Africa and the countries of Europe. The same route goes to America through Pacific Ocean. The work of the construction of a new seaport Bin Qasim near Karachi has also been completed. The work of establishing a new seaport at Gwadar in Balochistan is under progress which would promote Pakistan's sea trade with the world.

Air Routes

Apart from land and sea routes, the goods from Pakistan are also sent through air routes. Pakistan has trade relations with many countries of the world. Apart from the Muslim countries, trade takes place with European countries, China, Japan and America on a large scale. We also have trade relations with India, Sri Lanka, Singapore, Hong Kong and Australia.

Export Promotion Bureau

The government has established many institutions for the promotion of exports. Export Promotion Bureau is the most significant of all these institutions, which helps in solving the problems of traders and guides them in raising their exports for different countries. This bureau gives suggestions for increasing exports to various government and semi-government organizations. It sends trade delegations to different countries so that they could survey and assess the demand of Pakistani goods such as rice, cotton, cotton cloth, fish and fruits etc. For the promotion of trade in Pakistan, it makes agreements for Pakistan's participation in various International exhibitions and fairs. The important exports of Pakistan are rice, cotton, cotton yarn, cloth, hides and skin, sports goods, surgical goods, fish, prawns, fruits, etc.

Food Products

Amongst the food products rice is exported in the largest quantity and its export is increasing day by day. Pakistan grows the best quality rice, which is in great demand in the various countries of the world. Rice is specially exported to Saudi Arabia, Algeria, Iran, Kuwait, Dubai, Abu Dhabi,



Qatar, Yemen, Sri Lanka, Singapore, Indonesia, Britain, Canada, Italy and Germany. During 1998-1999 4673.8 thousand metric tons of rice were produced. By exporting the excess rice, Pakistan earned almost twenty billion rupees. Fish and prawns are caught in abundance from the coastal areas of Pakistan. They are either dried or tinned in boxes and sent to foreign countries. Fish and prawns are specially sent to Germany, America, Japan, France, Sri Lanka, Saudi Arabia, Jordan and some countries of Africa. They are also sent to Iran, Britain and some countries of Europe as well. Dry fruits are exported to Italy, Britain, America, Arab countries, Sri Lanka and Malaysia.

Cotton and Cotton Products

Cotton is the largest export of Pakistan. Yarn, thread, good quality cloth, T-shirts, socks, garments, towels are also prepared from cotton. Pakistan's cotton and cotton cloth has great demand in the various countries of the world. America, Germany, Italy, France, China, Japan, Australia, Poland, Canada, Saudi Arabia, Iraq, Morocco, Nigeria, Sri Lanka, India and Bangladesh import Pakistani cotton. In 1998-1999 the production of cotton in Pakistan was 8790.2 thousand bales.

Pakistan's cotton yarn and thread is also sent to different countries. It is exported to almost fifty countries of the world; among them the most significant are Japan, Italy, Germany, Britain, France, China, America, Iran, Iraq, Syria, Indonesia and many Arab and African countries.

Cloth made in Pakistan is also exported to many countries of the world, especially to Middle East and the Muslim countries of Africa, Sri Lanka and Malaysia. It is also exported to Italy, Germany, France and Britain.

Hides and Skin and Leather Goods

Pakistan also exports hides and skin in large quantity. The biggest buyers are France, Germany, Japan, Libya and Turkey. Hides and skin are also sent to Iraq, Iran, Arab countries and Britain. Leather goods are sent to France, Turkey, Spain, Britain, Germany, Saudi Arabia and Gulf States.

Electrical Goods

Electrical appliances are also made in Pakistan, which include household items like fans, bulbs, tube lights, cooler, etc. These goods are exported to Iran, Iraq, Saudi Arabia, various Gulf States, Sri Lanka, and Malaysia.

Pakistan also exports the heavy machinery used in the making of sugar, shoes, carpets, handicrafts, plastic goods, toys, goods made of glass and rubber.

These goods are also sent to the Muslim countries of Africa, which include Mauritania, Nigeria, Tanzania, Cameroon, Algeria, Malawi, Sierra Leone, etc.

Imports

Only those items are imported which are either not made in Pakistan or are imported for the purpose of industrialization in Pakistan. The government announces its import policy for a year, which includes the list of those commodities, which can be imported to Pakistan. The import of unnecessary items is mostly discouraged.

There are two ways for the payment of imported items. Generally cash payment is made for the import of goods. This payment is made in the foreign exchange, which Pakistan earns through its exports. Or else that money is utilized which Pakistan receives in the form of loans from other countries. Sometimes the loan, giving countries put this condition that in return of the loan given to them these countries should buy products from them only. With some countries trade also takes place by the exchange of goods for goods.

Machinery, oil, petroleum, tea and crude oil are the most important import items of Pakistan. For further industrial development Pakistan requires all kinds of heavy machinery. In Taxila Pakistan has established a big industry for making heavy machinery but still the country requires different kinds of machinery. Engines of trains and trucks and spare parts, machinery for making electricity, machines used in various factories, agricultural farms, offices and banks, television sets and machinery of similar kind is imported. Motorcars are mostly imported from Japan. Televisions are imported from Japan, Korea, Malaysia and other countries. Other machines, tractors and engines are bought from America, Canada, Germany, France, Britain, China, Japan, Italy and Korea.

Oil and Petroleum

Oil and petroleum is the basic need of present times. Pakistan imports oil, diesel oil and petrol from Saudi Arabia, Kuwait, Abu Dhabi, Iran and Iraq. The work for the search of oil has been made speedy in Pakistan. Even though recently large deposits of oil have been found but at the moment for making use of them a lot of work has to be done. Till 1999, the total production of oil in Pakistan was 10248 thousand barrel annually.

Importance of Energy

In today's world the development of any country depends on energy and its proper use. Different means are used for getting energy. The energy from sun, fire, fuel, air, and flowing water has always been used. But now we also get energy from

natural gas, coal, petroleum products, biogas, heat of the earth and nuclear resources for quick development. But due to the constant use of energy, its traditional sources are becoming less and less. Therefore, we have to think about finding out new sources as also for consuming energy wisely so that the present standard of living may be maintained.

The major principle for the conservation of energy is avoiding unnecessary use of energy. Some important measures are:-

- (i) To use energy for productive purposes only (all unproductive purposes should be avoided).
- (ii) To design buildings, factories and plazas, keeping in mind the basic principles of the conservation of energy.
- (iii) To make use of instruments that utilize less energy.

In short the available energy resources should be utilized properly, and the energy thus saved be utilized for the developmental projects.

Iron and Steel

Iron ore is excavated from mines. Steel is made by cleaning iron and mixing some other metals in the iron ore. In comparison to iron products, steel products are more strong, durable and better in use. The machines used in factories, in power houses and machines of different vehicles etc are mostly made of steel. This is the reason that the countries having iron and steel resources are industrially developed.

Due to shortage of good quality iron in Pakistan, steel industry could not be established. For important industries and electric houses, machinery generators etc are imported from Germany, Belgium, France, Britain, Italy, Japan, China and USA. In Pakistan a very big Steel Mill has been established with the cooperation of Russia in Karachi. Iron ore used in this industry is mainly imported from Australia. With the establishment of this Mills, the Steel requirements of Pakistan have been met to a greater extent. Big factories are established in Taxila and Kot Najeib Ullah for the manufacture of heavy machinery, generators and transformers etc. for different factories and electricity houses.

Due to the availability of steel in our country we have factories of ship building, defence goods, motorcars and railway engines, some other industries are also established.

Pakistan imports tea, powdered milk, good quality paper, some medicines, television sets, transistors, refrigerators, computers, watches, cameras and their films, different machines used in hospitals and weapons, etc. for defence.

Exercise

(A) Answer the following questions

1. State briefly the various means of transportation.
2. What is meant by means of communication? What are the important means of communication?
3. What is meant by import and export?
4. What are the important land and sea trade routes in Pakistan?
5. What are the important exports of Pakistan?
6. To which Middle Eastern countries are cotton and its products exported?
7. What are the main imports of Pakistan?
8. Discuss the importance of the means of transportation and communication in the promotion of trade.

(B) Fill in the blanks with correct words:

1. The resources, which are used for sending goods from one place to another, are called _____.
2. _____ invented telephone in 1876.
3. _____ is the most important invention of present times.
4. The Railway system of _____ is very famous in the world.
5. The buying and selling of goods among various countries is called _____.